

Derivative Chart Practice - Product and Quotient Rule

Directions: Use each table to find the indicated derivative. Show all work and simplify your answer.

1. $h(x) = f(x)g(x)$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-2	3	4	-1
2	5	-2	1	3
3	1	4	-3	2
6	-4	5	2	-2

$$h'(2) = f(2)g'(2) + g(2)f'(2) = (5)(3) + (-2)(1) = 15 - 2 = \boxed{13}$$

2. $h(x) = \frac{f(x)}{g(x)}$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-3	2	1
2	-2	4	-1	3
3	5	2	4	-2
6	1	-5	2	1

$$h'(3) = \frac{g(3)f'(3) - f(3)g'(3)}{[g(3)]^2} = \frac{(2)(4) - (5)(-2)}{2^2} = \frac{18}{4} = \boxed{\frac{9}{2}}$$

3. $h(x) = f(x)g(x) + x^2$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	4	-2	3	5
2	-3	6	2	-1
3	5	1	-2	4
6	2	-4	1	3

$$h'(x) = f(x)g'(x) + g(x)f'(x) + 2x$$

$$h'(1) = f(1)g'(1) + g(1)f'(1) + 2(1) = (4)(5) + (-2)(3) + 2 = \boxed{16}$$

4. $h(x) = \frac{f(x)}{g(x)} - 2x$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-5	2	1	4
2	3	-3	-2	1
3	4	5	3	-2
6	-2	4	5	1

$$h'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} - 2 \Rightarrow h'(6) = \frac{g(6)f'(6) - f(6)g'(6)}{[g(6)]^2} - 2$$

$$= \frac{(4)(5) - (-2)(1)}{4^2} - 2 = \frac{22}{16} - 2 = \boxed{-\frac{5}{8}}$$

5. $h(x) = 3f(x)g(x)$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-4	-3	2
2	-1	5	4	-1
3	6	2	1	3
6	3	-5	-2	4

$$h'(x) = 3[f'(x)g(x) + g'(x)f(x)] \quad h'(3) = 3[f'(3)g(3) + g'(3)f(3)]$$

$$= 3[(1)(2) + (3)(6)] = \boxed{60}$$

6. $h(x) = \frac{2f(x)}{g(x)}$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	-4	5
2	6	-3	2	1
3	-2	4	3	-1
6	5	1	-2	4

$$h'(x) = 2 \left[\frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} \right] \quad h'(1) = 2 \left[\frac{g(1)f'(1) - f(1)g'(1)}{[g(1)]^2} \right] = 2 \left[\frac{(2)(-4) - (3)(5)}{2^2} \right]$$

$$= \boxed{-\frac{23}{2}}$$

7. $h(x) = f(x)g(x) - x^3$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	3	2	-2
2	1	-5	4	3
3	6	2	-1	5
6	2	4	3	-1

$$h'(x) = f(x)g'(x) + f'(x)g(x) - 3x^2$$

$$h'(6) = f(6)g'(6) + f'(6)g(6) - 3(6^2) = (2)(-1) + (3)(4) - 108 = \boxed{-98}$$

8. $h(x) = \frac{f(x)}{g(x)} + x^2 - 4x$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	5	-2	1	4
2	-3	5	2	-2
3	4	1	-3	2
6	6	-4	1	3

$$h'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} + 2x - 4$$

$$h'(2) = \frac{g(2)f'(2) - f(2)g'(2)}{[g(2)]^2} + 2(2) - 4 = \frac{(5)(2) - (-3)(-2)}{5^2} = \boxed{\frac{4}{25}}$$

9. $h(x) = 2f(x)g(x) + 3x^2$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	3	-1	4
2	-5	1	3	-2
3	4	-4	2	1
6	1	5	-3	2

$$h'(x) = 2[f(x)g'(x) + f'(x)g(x)] \Rightarrow h'(1) = 2[f(1)g'(1) + f'(1)g(1)] + 6(1)$$

$$= 2[(2)(4) + (-1)(3)] + 6 = \boxed{16}$$

10. $h(x) = \frac{3f(x)}{2g(x)} - x^2$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	4	2	3
2	5	-2	-3	1
3	2	5	4	-1
6	-4	3	1	2

$$h'(x) = \frac{3}{2} \left[\frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} \right] - 2x \quad h'(3) = \frac{3}{2} \left[\frac{g(3)f'(3) - f(3)g'(3)}{[g(3)]^2} \right] - 2(3)$$

$$= \frac{3}{2} \left[\frac{(5)(4) - (2)(-1)}{5^2} \right] - 6 = \frac{3}{2} \left(\frac{22}{25} \right) - 6 = \boxed{-\frac{117}{25}}$$

11. $h(x) = f(x)g(x) + 4x^3 - x$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-3	2	-1
2	-2	4	5	3
3	1	2	-4	1
6	3	-5	2	4

$$h'(x) = f(x)g'(x) + f'(x)g(x) + 12x^2 - 1$$

$$h'(2) = f(2)g'(2) + f'(2)g(2) + 12(2^2) - 1 = (-2)(3) + (5)(4) + 48 - 1 = \boxed{61}$$

12. $h(x) = \frac{2f(x)}{g(x)} + x^3 - 3x^2$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	4	5	-2	1
2	-3	2	4	-1
3	6	-4	1	3
6	2	3	-5	2

$$h'(x) = 2 \left[\frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2} \right] + 3x^2 - 6x$$

$$h'(6) = 2 \left[\frac{g(6)f'(6) - f(6)g'(6)}{[g(6)]^2} \right] + 3(6^2) - 6(6) = 2 \left[\frac{(3)(-5) - (2)(2)}{3^2} \right] + 108 - 36$$

$$= 2 \left(\frac{-19}{9} \right) + 72 = \boxed{\frac{610}{9}}$$